

Problem 1 (Definitions). 1. *Define a set and give an example.*

2. *What is the cardinality of a set? Provide an example.*

3. *Explain the concept of a subset and provide an example.*

4. *Define the universal set and the empty set. Give examples.*

5. *What is the difference between an element and a subset?*

6. *Define the intersection of sets. Illustrate with an example.*

7. *Explain the concept of a power set. Provide an example.*

8. *Define the complement of a set. Give an example.*

9. *Differentiate between finite and infinite sets. Provide examples of each.*

10. *Define the Cartesian product of sets. Illustrate with an example.*

Problem 2 (Types of sets). 1. *Define and provide an example of a finite set.*

2. *Explain the concept of an infinite set and provide an example.*

3. *Define a singleton set and give an example.*

4. *What is the difference between equal sets and equivalent sets?*

5. *Define and provide examples of a null set, a universal set, and a subset.*

6. *Explain the concept of a well-defined set. Give an example.*

7. *Define a proper subset and give an example.*

8. *What is a countable set? Provide an example.*

9. *Define and provide examples of disjoint sets.*

10. *Explain the concept of a power set and give an example.*

Problem 3 (Venn diagram). 1. *Draw a Venn diagram illustrating the intersection of two sets.*

2. *Create a Venn diagram for three sets illustrating a proper subset.*

3. *Illustrate the complement of a set using a Venn diagram.*

4. *Draw a Venn diagram representing the union of two sets.*

5. Create a Venn diagram for the symmetric difference of two sets.
6. Illustrate the Cartesian product of two sets using a Venn diagram.
7. Draw a Venn diagram representing the power set of a given set.
8. Create a Venn diagram for the intersection of three sets.
9. Illustrate the union of three sets using a Venn diagram.
10. Draw a Venn diagram representing the complement of the union of two sets.
11. Use Venn diagrams to investigate the truth or falsity of each of the following for sets $A, B, C \subset U$.
 - $A \Delta (B \cap C) = (A \Delta B) \cap (A \Delta C)$.
 - $A - (B \cup C) = (A - B) \cup (A - C)$

Problem 4 (Set operations). 1. Express the intersection of sets A and B in set-builder notation.

2. Write the union of sets X and Y in roster form.
3. Perform the Cartesian product of sets M and N .
4. Calculate the complement of set P .
5. Express the symmetric difference of sets Q and R in set-builder notation.
6. Write the intersection of three sets, A , B , and C , in roster form.
7. Perform the union of four sets, X , Y , Z , and W .
8. Calculate the power set of set S .
9. Express the complement of the union of sets P and Q in set-builder notation.
10. Perform the intersection of sets U and V using Venn diagrams.

Problem 5 (Properties of set operations). 1. List 9 properties of unions and intersections of set understand how to prove them.

2. List 8 properties of set difference and understand how to prove them.
3. If $A = [-3, 0]$, $B = [-2, 5]$ with $U = \mathbb{R}$ determine: $A \cup B$, $A \cap B$, A^C , $A \Delta B$, $A - B$, $B - A$
4. Use properties of set operations to investigate the truth or falsity of each of the following for sets $A, B, C \subset U$.
 - $A \Delta (B \cap C) = (A \Delta B) \cap (A \Delta C)$.

- $A - (B \cup C) = (A - B) \cup (A - C)$

Problem 6 (Cartesian products). 1. Let $A = \{1, 2\}$ and $B = \{a, b\}$. Find $A \times B$.

2. Consider sets $X = \{a, b, c\}$ and $Y = \{1, 2\}$. Determine $X \times Y$.

3. If $M = \{x, y\}$ and $N = \{1, 2, 3\}$, find $M \times N$.

4. Let $A = \{1, 2, 3, 4\}$, $B = \{4, 2, 3\}$ and $C = \{2, 5\}$ determine:

$$A \times B, B \times A, (A \cup B) \times C$$

5. Let $G = \{1\}$ and $H = \{x, y, z\}$. Find $G \times H$.

6. Consider sets $I = \{a\}$ and $J = \{1\}$. Determine $I \times J$.

Problem 7 (Practical problems). 1. Let $A = \{1, -1\}$ and $B = \{x | x \in \mathbb{Z} \text{ and } x^2 - 1 = 0\}$ prove that $A \subset B$

2. In a survey of 100 students, 60 liked Mathematics, and 40 liked both Mathematics and Science. How many students liked only Science?

3. A deck of cards contains 52 cards. If a card is drawn at random, what is the probability of drawing a face card (jack, queen, or king)?

4. Consider two sets: $A = \{2, 4, 6, 8\}$ and $B = \{3, 6, 9\}$. Find the union and intersection of these sets.

5. In a group of 50 people, 30 like coffee, and 20 like tea. If 10 people like both coffee and tea, how many people do not like either beverage?

6. If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, find the Cartesian product $A \times B$.

7. Solve the equation $2x + 3 = 7$ and represent the solution set using set notation.

8. A company produces two types of products: Product A and Product B. The set of customers who bought Product A is $A = \{1, 2, 3, 4, 5\}$, and for Product B, it is $B = \{4, 5, 6, 7\}$. Find the set of customers who bought at least one of the products.

9. If $|P| = 5$ and $|Q| = 3$, what is the cardinality of the union of sets P and Q if $|P \cap Q| = 2$?

10. A school offers three language courses: English, French, and Spanish. Out of 120 students, 75 are taking English, 60 are taking French, and 45 are taking Spanish. If 30 students are taking both English and French, 15 are taking both French and Spanish, and 20 are taking both English and Spanish, how many students are taking all three languages?

11. Consider a set X with 8 elements. How many subsets does X have, including the empty set and X itself?

Problem 8 (Definitions). 1. Define a relation between two sets. Provide an example.

2. Explain the concepts of domain and range in the context of relations.

3. Define an ordered pair. How is it related to relations?

4. Explain what it means for a relation to be reflexive. Give an example.

5. Define an equivalence relation. Provide an example.

6. What is the difference between a function and a relation?

7. Define a one-to-one (injective) relation. Give an example.

8. Explain the concept of a many-to-one (non-injective) relation. Provide an example.

9. Define a symmetric relation. Give an example.

10. Explain what it means for a relation to be antisymmetric. Provide an example.

Problem 9 (Types of Relations). 1. Define and provide examples of the following types of relations: reflexive, symmetric, transitive.

2. Explain the concept of an equivalence class in the context of equivalence relations.

3. Define and provide examples of the following types of relations: partial order, total order.

4. What is the difference between a reflexive and an irreflexive relation? Give examples.

5. Define an asymmetric relation. Provide an example.

6. Define and provide examples of the following types of relations: finite, infinite.

7. Explain the concept of a well-defined relation. Give an example.

8. What is a transitive closure of a relation? Provide an example.

Problem 10 (Functions definitions). 1. Define a function from set A to set B . Provide an example.

2. Explain the concepts of domain, codomain, and range in the context of functions.

3. Define the terms "injective," "surjective," and "bijective" in the context of functions.

4. Explain what it means for a function to be one-to-one. Give an example.

5. Define a composite function. Provide an example.

6. What is the difference between an onto and a into function? Give examples.

7. Define a constant function. Provide an example.

8. Explain the concept of an inverse function. Provide an example.
9. Define a piecewise function. Give an example.
10. Explain what it means for a function to be even or odd. Provide examples.

Problem 11 (Properties). 1. Define and provide examples of the following properties: reflexive, symmetric, transitive.

2. Explain the concept of a reflexive closure of a relation. Provide an example.
3. Define the terms "left-unique" and "right-unique" in the context of functions.
4. Explain what it means for a function to be a bijection. Give examples.
5. Define an identity function. Provide an example.
6. Explain the concept of a surjective function. Give an example.
7. Define a recursive function. Provide an example.
8. Explain what it means for a function to be continuous. Give examples.
9. Define a monotonic function. Provide an example.
10. Explain the concept of an onto function. Give examples.

Problem 12 (Composition). 1. Given functions $f(x) = 2x$ and $g(x) = x^2$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

2. If $f(x) = 3x - 1$ and $g(x) = \sqrt{x}$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.
3. Define the composition of three functions and provide an example.
4. If $h(x) = x^2 + 1$ and $k(x) = \frac{1}{x}$, find $(h \circ k)(x)$ and $(k \circ h)(x)$.
5. Given $f(x) = 2x$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Problem 13 (Practical Problems). 1. A company charges a fixed fee of 50 dollars plus 30 dollars per hour for its services. Write a function representing the total cost in terms of hours of service.

2. In a town, the population is modeled by the function $P(t) = 1000e^{0.02t}$, where t is the time in years. Find the population after 5 years.
3. A car rental company charges 30 dollars per day plus 0.25 dollars per mile driven. Write a function representing the total cost in terms of the number of miles driven in a day.
4. The height of an object in free fall can be modeled by the function $h(t) = 4.9t^2$, where t is the time in seconds. Find the height after 3 seconds.

5. Express the following real-world scenario as a function: The cost of printing x copies of a book is given by $C(x) = 5x + 100$ dollars.

6. Find the inverse of :

- $y = mx + c$ where $m, c \in \mathbb{R}$
- $y = \exp x + 2$
- $y = 3x^3 - 5$